# SHIVAM LALAKIYA

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## Professional Summary

- Data Scientist with **4+ years** of experience building analytics that reach production and stay there: propensity and forecasting models deployed into live CRM operations, unattended data quality and reporting pipelines serving multiple teams, and LLM applications on secure enterprise infrastructure.

## Experience

**Washington University in St. Louis** **Feb 2024 – Present**

*Data Analyst, Grateful Patient Program* *St. Louis, MO*

- Built an internal **AI analytics assistant** (**Next.js**, **LangGraph**, **MCP**, vector retrieval) on the university's secure AI gateway, indexing live **Salesforce** data and internal GitHub repositories to answer analytical questions in seconds; shipped from a standing start in **10 weeks**.
- Developed and productionized donor **propensity models** achieving a **66x lift** at the top of the ranked list, deployed into the CRM as interest codes now driving live gift-officer targeting alongside scoring and tiering frameworks adopted across Prospect Management and Research.
- Architected an automated **data quality pipeline** (**SQL Server Agent**, **16** checks across **11** CRM objects) serving **4** teams at **$0** infrastructure cost, with **519** runs over **17** failure-free weeks cutting missing contact preferences **73%** and unlinked accounts **79%**.
- Engineered **fuzzy matching** and duplicate-detection engines scanning a **703K-row** contact table weekly, surfacing **2,100+** probable duplicate pairs per cycle and unifying external VSE records with the CRM.
- Architected a multi-model cash-flow forecasting suite (**ARIMA, LSTM, Prophet**) over **26** fiscal years of actuals with macroeconomic inputs and multi-tier recession stress-testing, delivering **5-year** projections to Treasury and the **Board of Trustees**.
- Automated **8** recurring production reporting systems and **100+** custom reports across **50** departments, collapsing a **4-hour** manual cycle to **under 5 minutes** with zero manual rebuilds.

**Invicro, LLC.** **Sept 2023 – Feb 2024**

*Associate Data Manager* *Boston, MA*

- Engineered an automated data pipeline with **Apache Airflow** for pathological image transfer, reducing processing time by **20%** and ensuring on-time delivery for clinical research.
- Deployed **computer vision** models for pathological image analysis, achieving a **10%** improvement in bio-marker identification and contributing to a **15%** increase in study sponsors.
- Architected data transformation models in **dbt**, streamlining workflows and driving a **25% reduction** in downstream reporting errors.

**Ring Therapeutics** **May 2022 – Dec 2022**

*Data Science Co-op* *Boston, MA*

- Leveraged **NLP** (positional encoding) and **Graph Neural Networks** to predict gene tropism at a **0.75 F1-score**, generating vectors for targeted cell therapy delivery.
- Fine-tuned a **Protein-BERT** model on proprietary Anello sequences, reaching a **0.8 F1-score** for tropism classification, and automated **90%** of data loading via **Dockerized Airflow** DAGs.

## Technical Skills

**Languages:** Python, SQL (T-SQL), R, Shell Scripting, Java, MATLAB
**ML & AI:** Forecasting (ARIMA, LSTM, Prophet), Propensity Modeling, NLP & Transformers (BERT), LLM Applications (RAG, LangGraph, MCP, Agents), Computer Vision
**Data Engineering:** ETL, Apache Airflow, dbt, SQL Server Agent, Docker, AWS, Kafka, REST APIs, Git, CI/CD
**Platforms & BI:** Salesforce, SQL Server, Tableau, Power BI, React/Next.js

## Education

**Northeastern University, Boston, MA** **Sept 2021 – July 2023**

*Master of Science in Data Analytics Engineering* *GPA: 3.8/4.0*

**Sardar Vallabhbhai National Institute of Technology (NIT Surat), India** **July 2017 – May 2021**

*Bachelor of Technology in Electronics and Communication Engineering* *GPA: 8.1/10*

## Open Source

- **PhilanthroPy** (author & maintainer): **scikit-learn**-style donor analytics library for nonprofit fundraising (donor scoring, lifecycle segmentation, campaign metrics). **v0.10.0**, **265** PyPI downloads/mo, **22** contributors, **3,082** tests.
- **28 merged pull requests** to **LlamaIndex** (52K stars), **sktime** (10K), **MAPIE**, **PyMC-Marketing**, and **TorchCP**, spanning memory backends, vector-store metadata, and conformal prediction.